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Who faces travel constraints? Analyzing household and individual travel challenges in automobile-dependent central Texas

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Highlights

- Develops a multidimensional framework integrating household affordability with intra-personal travel constraints across health, safety, comfort, accessibility, and time dimensions.
- Households with high housing and travel cost burdens and elevated intra-personal travel constraints, especially low-income ones, exhibit suppressed travel demand.
- Finds that single-mother households, primarily Black and Hispanic/Latino, experience disproportionately pronounced intra-personal travel constraints.
- Suggests policy interventions to expand non-automobile options, implement affordability programs, and provide mobility support tailored to the specific needs of certain populations.

Abstract

Understanding travel constraints is essential to evaluating whether individuals can meet daily transportation needs. Prior studies often rely on single indicators, such as car ownership or travel costs. This study develops a

multidimensional framework that integrates a household-level Housing and Transportation Cost-to-Income Ratio (HT CIR) with an individual-level Intra-personal Travel Constraints (ITC) propensity model. Drawing on regional housing and travel data merged with the Next-Generation National Household Travel Survey (NextGen NHTS), this study classifies household types in Central Texas by constraint profiles.

Households with high HT CIR face affordability challenges, while those with high ITC propensities encounter barriers related to health, safety, comfort, accessibility, and time. Findings show that high HT CIR and lower income suppress travel demand, particularly among households with high ITC propensities. Social and demographic factors—including limited income, insufficient car ownership, limited English proficiency, single-parent status, race, and retirement—increase the likelihood of high ITC propensities. Notably, single-mother households displayed distinct patterns of constraint, with a larger share experiencing health, comfort, affordability, accessibility, and time-related challenges compared with other household types. These disparities are compounded for Black and Hispanic/Latino single mothers, underscoring the intersection of race and gender in shaping mobility barriers.

Policy implications vary by context. In areas with both high HT CIR and high ITC, interventions should prioritize alternatives to car use, affordability programs, and mobility services targeted at specific populations. Where high HT CIR dominates, financial support is most urgent. Expanding non-automobile access is critical to reducing car dependence and mitigating inequities across diverse populations.

Introduction

In transportation planning, a key consideration is whether individuals can use the transportation system to meet their daily needs, which include activities such as attending meetings, accessing medical services, shopping, visiting family, and recreational pursuits (Ahern and Hine, 2012; Hjorthol, 2013; Musselwhite and Haddad, 2018; Nordbakke and Schwanen, 2015; Siren et al., 2015; van Wee and Mokhtarian, 2023). However, several travel constraints can hinder access to transportation, including high costs, limited options, and physical disabilities. These constraints can prevent individuals from fully participating in social, political, and cultural life (Luiu et al., 2017; Palm et al., 2021).

Travel constraints may vary depending on context, where what may appear as a limitation in one context could be a deliberate choice in another. For instance, high travel costs may not constitute a constraint if individuals willingly accept them, such as when they have sufficient income (Hancock, 1993). However, these costs become constraints when individuals are forced to bear them due to a lack of alternative transportation options. Similarly, limited transportation infrastructure might not pose a challenge if individuals choose to live in a region for other benefits like affordable housing, community culture, or proximity to work.

In the United States, the increase in car use over the past five decades has led to a shift toward automobile-dependent transportation patterns. These patterns have resulted in excessive reliance on cars and longer driving distances (Handy et al., 2005; Wachs, 1984). However, car use is not always a voluntary choice. The lack of alternative mobility options often forces individuals to rely on cars and exacerbates mobility difficulties for those who cannot drive (Brown, 2017; Zivarts, 2024).

Existing literature on travel constraints examines external factors, such as limited transportation options and financial resources, and intrapersonal factors, including psychological perspectives and travel preferences (Crawford et al., 1991). While planners generally view high travel costs and limited infrastructure as constraints, these factors are not always burdensome since some individuals may accept them based on personal preferences or lifestyle choices. Current studies are limited by using a single criterion, such as car ownership or travel costs, to

assess constraints. Improving transportation policies demands an understanding of the diverse needs of various population groups, considering both external determinants—such as socioeconomic conditions, affordability, and regional context—and intrapersonal factors, including individual attitudes and preferences (Crawford et al., 1991).

This study aims to address these gaps by analyzing national survey data to identify households affected by diverse travel constraints, with a focus on the Greater Austin area, a region heavily reliant on cars. It first introduces a housing and transportation cost index to categorize household-level affordability constraints, and develops a model to estimate individuals' propensity to experience intrapersonal travel constraints—defined as the probability of encountering personal or situational barriers to reaching desired destinations within the existing transportation system. Subsequently, this study examines the travel behaviors associated with travel constraints in households across five counties in central Texas. By introducing a novel measure of travel constraints, this study seeks to inform policies that more effectively address the diverse needs and challenges of populations facing different types of travel constraints.

Section snippets

Measuring travel constraints at the household level

Planning literature measuring populations' travel constraints primarily focuses on external and intra-personal factors. External constraints include the lack of available transportation options and limited financial resources. For instance, low-income populations with cars face greater financial burdens than their carless counterparts, especially when pedestrian-friendly environments or robust public transit services are lacking (Banister, 1994; Currie and Senbergs, 2007). Intra-personal ...

Individual-level travel constraints

Individuals' ability to travel is not uniformly influenced by factors such as the built environment and available modes of transportation. Additionally, individuals may encounter disproportionate travel barriers associated with their social status, psychological condition, or physical health. Empirical studies highlight that individuals can experience travel constraints in accessing their preferred travel modes, depending on their personal judgment regarding health concerns, safety and ...

Methodology

The selected region for this study, the Austin metropolitan area in Texas, USA, exhibits travel patterns similar to those of other mid-sized to large cities across North America. In these cities, the prevalent reliance on private vehicles as the primary mode of transportation is accompanied by evident gaps in investment in both transit infrastructure and active transportation. Data from the 2023 American Community Survey (ACS) five-year estimates indicate that Austin's drive-alone rate is 62%, ...

Factors that influence ITC propensity

In this study, ITC propensity refers to the probability that a household member perceives travel constraints during daily trips within the transportation system. In other words, it measures the likelihood that a household has members who experience intrapersonal travel barriers. For example, low ITC propensity indicates a lower

probability that household members perceive constraints in maintaining their daily trips and vice versa. The ITC propensity model results presented in Table 4 reveal ...

Policy implications and conclusions

This study measures household-level affordability and the propensity for intra-personal travel constraints in central Texas to identify the four levels of travel constraints in an automobile-dependent region representative of North America. According to the literature benchmark, household-level travel affordability constraints were quantified as the share of HT costs in income, with an HT CIR exceeding 45%, indicating unaffordable expenses. Intrapersonal travel constraints were quantified ...

Limitations and further research

Although this study significantly contributes to understanding travel constraints by examining both household-level and personal-level factors using available travel behavior survey data, it has limitations related to available data and aggregation bias. The ITC propensity scores, based on questions in the NextGen NHTS, primarily address specific aspects of personal concerns that lead to the trip reduction or cancellation, such as high travel costs, discomfort, and disability. However, ...

Declaration of competing interest

The author(s) declare no conflicts of interest. ...

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